Boundaries of Differentiated Product Markets and Retailer Pricing

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Abstract

This paper studies the effects of misspecified boundaries of competition on optimal retail pricing using store-level supermarket scanner data. We focus on two types of misspecification: (i) misspecification of the demand estimation problem arising from overly restrictive functional forms or the omission of relevant goods; (ii) misspecification of the retailer's decision problem, in which prices are set separately rather than jointly across related categories. Quantifying the costs associated with either form of misspecification is challenging because it requires a flexible yet valid demand system. We take a nonparametric approach to estimating demand that imposes minimal restrictions on cross-price effects, satisfies properties required by economic theory, and nests standard parametric models. In our empirical analysis, we provide descriptive evidence across nine product groups that cross-category effects are large and often nuanced in their sign and magnitude. We then zoom in on refrigerated juices and estimate multi-category demand systems with varying degrees of flexibility. We document and explain differences in elasticity estimates across specifications, and solve for optimal prices under each model to quantify losses in profits due to misspecification.

Keywords: multi-category demand, nonparametric demand estimation, Bernstein polynomials, model misspecification, omitted variable bias

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1 Introduction

Delineating boundaries of product competition is at the center of many questions in marketing and economics. Yet in most empirical work, such boundaries already begin to take shape prior to the analysis. For example, consider two key inputs into any demand analysis: a subset of goods to represent the consumer choice set, and a stochastic specification of demand to take to the data. Decisions about the former implicitly restrict substitution between the included goods and the remaining goods excluded from the analysis, while decisions about the latter can also restrict the scope of substitution through various functional form and/or parametric assumptions. As a consequence, questions of misspecification often linger when estimating demand for differentiated products. Is the set of goods defined broadly enough in order to capture all relevant dimensions of substitution? Is the model flexible enough to capture this substitution? How will misspecification in the estimation problem impact subsequent policy prescriptions?

The goal of this paper is to quantify the costs of misspecified boundaries of competition on optimal retailer pricing policies. Our specific target is the loss in profits accrued by a retailer who solves for optimal prices under two possible forms of misspecification. The first is a misspecification of boundaries in the *estimation problem* which arises when (i) the demand model is overly restrictive and, for example, only allows for strict substitution in a market with complementarities;[1] or (ii) the market is defined to be too narrow and omits relevant goods from the demand system. In either case, the concern is that estimates of key demand derivatives will be biased. The second is a misspecification of boundaries in the retailer's *decision problem* which arises when the retailer sets prices separately, rather than jointly, across categories. The scope of a retailer's pricing problem is at the center of many questions related to category management (Basuroy et al., 2001) and the organization of supply (Kadiyali et al., 2000; Sudhir, 2001). For example, both retailer profits and the degree of market power have been shown to depend on whether a retailer sets prices jointly across categories and internalizes cross-category substitution (Smith and Thomassen, 2012; Thomassen et al., 2017; Ershov et al., 2021).

Our empirical context is pricing consumer packaged goods (CPGs) using store-level grocery retail data. We focus on estimating demand within product groups (e.g., salty snacks) spanning multiple related categories (e.g., potato chips and pretzels). Quantifying the costs of misspecification in this empirical context is challenging because we need estimates of substitution patterns that are flexible — in terms of the sign and magnitude of demand derivatives — but also obey certain properties of microeconomic theory to ensure valid counterfactual profit predictions. Workhorse demand models for differentiated products such as BLP (Berry et al., 1995) are micro-founded but relatively inflexible for our purposes, in the sense that functional forms constrain cross-price derivatives to be positive. In contrast, many regression or machine learning-based models are relatively flexible but often lack micro-foundations (Berry and Haile, 2021).

[1] Because the retailer's pricing problem only depends on derivatives of the demand function, we remain agnostic towards the source of negative cross-price effects and whether they reflect true complementarity in consumption or simply co-purchases.

To overcome these challenges, we take a nonparametric approach to demand estimation. We first specify a demand system in which market quantities are expressed as general functions of prices, product characteristics, and (endogenous) structural unobservables. We then adopt the nonparametric estimator of Compiani (2021), which uses Bernstein polynomials to approximate the system of inverse demand equations. A nonparametric approach is attractive for our purposes because it does not restrict the sign of the cross-price effects. In particular, we only require the underlying demand function to satisfy minimal identification restrictions (Berry and Haile, 2014), which are micro-founded but allow for a range of substitution/complementarity patterns. Moreover, the advantage of Bernstein polynomials is that they are amenable to imposing relevant constraints such as own-price monotonicity (i.e., downward-sloping demand). We can also impose other constraints such as strict substitution in order to derive less flexible but nested models for comparison.

Our empirical analysis uses store-level transaction data from one major grocery retail chain in the United States. We provide two main sets of results. The first set is descriptive: we fit a battery of log-log models to data from nine broad product groups to show that cross-category effects are appreciably large, and so a more structural approach to investigating misspecification is warranted. In doing so, we also find that some cross effects are easier to sign a priori than others. For example, the majority of cross effects in the Baking Goods or Detergents/Softener groups are negative, which is consistent with our expectations and existing empirical work on cross-category complements (Manchanda et al., 1999; Song and Chintagunta, 2006). In other groups like Refrigerated Juice, estimated cross effects are much more dispersed and so the nature of substitution is harder to characterize a priori. This would ordinarily complicate the specification of a structural model of demand and provides further motivation for our flexible nonparametric approach.

Our second set of results is structural and zooms in on the demand for five product categories in the Refrigerated Juice product group. We take two versions of our nonparametric estimator to the data: a flexible specification that imposes no constraints on the sign of cross-price derivatives and a restricted specification that constrains cross-price derivatives to be positive. We find that the flexible estimator uncovers a mix of positive and negative cross-price elasticities. Further, it delivers a better out-of-sample fit than the more restricted model. We then use the flexible specification to quantify biases in estimated elasticities that result from omitting relevant goods from the demand system. Specifically, we estimate a sequence of smaller models in which we iteratively remove one category from the demand system, and then compare estimated elasticities to the "true" elasticities from the full model that includes all five categories. We find that the direction of the bias in the smaller, misspecified models is consistent with standard omitted variable bias intuition. That is, the sign of the bias is governed by both (i) the sign of the omitted elasticity; and (ii) the correlation in prices between the included price (or its instrument) and the omitted price(s). While most recent work on large-scale demand estimation emphasizes (i) alone, our results suggest that checking (ii) is just as important.

Finally, we quantify the costs of misspecification through a pricing counterfactual. We solve for optimal retail prices and profits under both flexible and substitutes nonparametric systems

discussed above, as well as a third system that omits cross-category effects altogether. We treat the latter two specifications as misspecified. We also solve for two types of pricing policies that have different implications for the boundaries of the retailer's pricing problem: a "joint pricing" policy where the retailer maximizes total profits across all juice categories, and a (misspecified) "isolated pricing" policy where the retailer separately maximizes category profits. Relative to the case of joint pricing with a flexible model, profits are 6.0% lower when demand restricts cross effects to be positive but prices are solved jointly; 22.7% lower when demand is flexible but prices are solved separately in each category; 17.7% lower when cross effects are restricted to be positive and prices are solved separately; and 28.4% lower when cross effects are omitted altogether. A conduct test reveals that the retailer's current practices in Refrigerated Juice are closest to "isolated pricing," which further motivates the pricing boundaries dimension of our profit decomposition.

Our results suggest that the costs of misspecification are substantial. We find that misspecification of the decision problem is first-order relative to misspecification in the estimation problem as the worst-case profits under joint pricing are still higher than the best-case profits under isolated pricing. Given that our misspecified demand systems are still nonparametric, our results likely provide a lower bound to the losses in retailer profits due to misspecification of the estimator. In other words, a more restrictive (parametric) model that assumes substitution among all goods may lead to larger profit losses.

Contributions to the Existing Literature

Our work contributes to a large literature on pricing differentiated products. Much existing work has focused on pricing relatively narrow assortments of imperfect substitutes, such as varieties of cereal (Nevo, 2001), ketchup (Besanko et al., 2003), milk (Handel et al., 2013), orange juice (Montgomery, 1997; Chintagunta et al., 2003; Nair et al., 2005; Dubé et al., 2008, 2010), and yogurt (Vilcassim and Chintagunta, 1995; Kim et al., 2002; Draganska and Jain, 2006; Pavlidis and Ellickson, 2017). While focusing on a single category can simplify the analysis, one potential concern is that substitution patterns will be distorted if related goods outside of the defined category are omitted from the system.[2] For example, when modeling the demand for orange juice alone, all remaining juice flavors would enter into an outside good typically assumed to be weakly separable from the inside goods (Chintagunta and Nair, 2011). Smith et al. (2019) show that demand for juice is generally not separable by flavor, calling into question the simple functional relationships commonly assumed between the inside and outside goods. More broadly, this raises questions of the scope of a product market when solving for optimal prices and, in turn, what a misspecified model implies for optimal prices and profits.

[2]Note that in CPG markets, academic researchers typically adopt the category definitions given by data providers like IRI and Nielsen. These categories may not align with the category definitions used by retailers, however. For example, refrigerated orange juice is often treated as a category in academic research, but the broader set of refrigerated juices could constitute a category for the retailer (in the sense that the retailer coordinates marketing decisions for all juice products, not just orange juices). Our goal is to document the extent of substitution outside of typical category definitions used in academic research.

Pricing wider product assortments has been made possible with the rise of multi-category models of demand (see reviews in Berry et al., 2014 or Dubé, 2019). However, most empirical applications have focused on categories where the nature of substitution is clear a priori. Examples include the complementarity between cake mix and cake frosting (Manchanda et al., 1999; Ma et al., 2012), laundry detergent and fabric softener (Song and Chintagunta, 2006, 2007; Mehta, 2007), or milk and cereal (Lee et al., 2013). Moreover, most structural multi-category demand models are developed at the individual-level and require data on household choices (e.g., Song and Chintagunta, 2007; Mehta, 2007; Thomassen et al., 2017; Lewbel and Nesheim, 2019). Our paper contributes to this literature by providing a fully structural yet flexible model of aggregate demand that remains suitable when the focal market spans multiple categories and consists of products with nuanced patterns of substitution/complementarity.

Our paper also contributes to a fast-growing literature on flexible demand estimation. In this literature, a few distinct approaches have emerged to accommodate more flexible substitution patterns and wider product assortments. One approach is to extend the framework of BLP using more flexible specifications of inverse demand functions (Fosgerau et al., 2021; Monardo, 2021) or distributions on the random coefficients (Fox et al., 2016; Wang, 2021b). A second approach is to reformulate the logit discrete choice assumption as discrete choice over product bundles (Iaria and Wang, 2020; Wang, 2021a; Ershov et al., 2021), which allows for joint consumption and product complementarities. A third approach is to estimate demand functions, or key derivatives of those functions, nonparametrically (Blundell et al., 2017; Wang and Huang, 2019; Compiani, 2021). A final approach is to exploit flexible machine learning methods, such as embedding models (Kumar et al., 2020; Chen et al., 2020), matrix factorization (Donnelly et al., 2021; Ruiz et al., 2020), and deep learning (Gabel and Timoshenko, 2021).

While each of the estimation frameworks outlined above offers some aspect of flexibility, we believe our nonparametric approach, based on the estimator in Compiani (2021), is most suitable for our research questions. Quantifying the costs of misspecification on retailer pricing requires a "ground truth" model that guards against functional form misspecification, which is a virtue of any nonparametric approach. Our framework also makes it possible to impose theory-based restrictions that discipline the estimator and enforce key regularity properties on the profit function used in our pricing counterfactuals. Finally, our framework nests more restrictive models — such as the specification where all products are substitutes — which allows us to cleanly compare models and quantify the incremental value of model restrictions. One drawback is that our estimator will be subject to a curse of dimensionality that will make it infeasible to estimate demand for very large assortments. That said, we believe our results provide a valuable first step in demonstrating that misspecification matters for optimal pricing, even in a just few related categories, which should in turn motivate more work on a larger scale in this area.

Finally, while our methodological approach follows closely from Compiani (2021), our analysis is different in the following ways. First, the nonparametric estimation methods are used in service to key policy questions about market definition and boundaries of competition. Doing so requires us to

conceive of, operationalize, and then estimate a host of misspecified models, which is unique to our analysis. Our empirical analysis is also broader in scope. Compiani (2021) estimates demand for two goods in one category (regular/organic strawberries), and finds evidence of strict substitution. We apply the nonparametric estimator to data spanning five categories and find evidence for both substitution and complementarity. Finally, while Compiani (2021) uses estimates to quantify the pass-through of a tax and the "portfolio effect" of multi-product pricing, our counterfactuals focus on a comparison of retailer profits from pricing under different sources of misspecification.

The remainder of this paper is organized as follows. Section 2 outlines the focal demand system and Section 3 presents the nonparametric estimator. Section 4 reports results from a variety of simulation studies to illustrate the flexibility of the proposed method as well as the scope for the effects of boundary misspecification on price elasticities. Section 5 describes the retail scanner data used in our empirical analysis. Section 6 presents descriptive results on cross-category elasticities and price correlations across a broad set of nine product groups. Section 7 presents results from our nonparametric analysis of demand and subsequent pricing counterfactuals. Section 8 concludes.

2 A Flexible Model of Demand

Consider a continuum of consumers belonging to many markets, where each market $t = 1, \ldots, T$ is defined by a fixed assortment of goods $j = 1, \ldots, J$ sold at prices $\mathbf{p}_t = (p_{1t}, \ldots, p_{Jt})$, a set of observed product/market characteristics $\mathbf{x}_t$, and a vector of unobserved product characteristics $\boldsymbol{\xi}_t = (\xi_{1t}, \ldots, \xi_{Jt})$. Aggregate demand can then be expressed as:

$$q_{jt} = \sigma_j\big(\mathbf{p}_t, \mathbf{x}_t, \boldsymbol{\xi}_t\big), \tag{1}$$

where q_{jt} is the demand of good j in market t and $\sigma_j(\cdot)$ is the demand function for good j. The range of σ_j can accommodate cases where the dependent variable represents market shares (discrete choice) or quantities (continuous choice). Specifying a model for quantities is important for at least two reasons. First, the assumption that each consumer buys at most one unit from the choice set is sometimes likely to be violated. For example, consumers might purchase several units of multiple different goods (Kim et al., 2002; Dubé, 2004). This is especially important in the presence of complementary products, which inherently violate the substitution patterns implied by standard discrete choice models. Second, even if the products are substitutes, the data available to researchers often only contain information about quantities as opposed to market shares. In order to apply standard models, the researcher typically needs to take a stance on the size of the market and use that to convert quantities into shares. This then raises questions about how robust the results are to the assumed market definition. In contrast, the approach discussed here does not require any such assumptions.

As is typical in the demand estimation literature, we modify (1) to introduce the notion of a product index. First, partition $\mathbf{x}_t = (\mathbf{x}_t^{(1)}, \mathbf{x}_t^{(2)})$ with $\mathbf{x}_t^{(1)} = (x_{1t}^{(1)}, \ldots, x_{Jt}^{(1)})$ and let each $x_{jt}^{(1)} \in \mathbb{R}$

(the model trivially extends to the case where $x_{jt}^{(1)}$ is a vector). Then define the index:

$$\delta_{jt} = \beta_j x_{jt}^{(1)} + \xi_{jt} \tag{2}$$

which requires that $x_{jt}^{(1)}$ and ξ_{jt} only affect demand through δ_{jt}. However, the way in which δ_{jt} itself enters the demand function is left unrestricted (as is the case with $\mathbf{x}_t^{(2)}$ and $\mathbf{p}_t$). This index structure aids in nonparametric identification and, as shown in Berry and Haile (2021), is implicitly satisfied in many common parametric demand models. One example is a mixed logit model where the coefficient on $x_j^{(1)}$ is non-random. Given (2), we can rewrite the demand system as:

$$q_{jt} = \sigma_j \left(\boldsymbol{\delta}_t, \mathbf{p}_t, \mathbf{x}_t^{(2)} \right) \tag{3}$$

which will serve as the target estimand of our nonparametric approach.

The nonparametric identification of the demand system above was first outlined in Berry et al. (2013) and Berry and Haile (2014), and then summarized more recently in Berry and Haile (2021). The main econometric challenge is that the demand for each good is a function of the entire J-vector of indices $\boldsymbol{\delta}_t$, which means that the right-hand-side of (3) contains not one but J unobservables $\boldsymbol{\xi}_t = (\xi_{1t}, \ldots, \xi_{Jt})$. In this sense, the equations defined by (3) are not standard regression equations, which in turn complicates identification and estimation.

A solution to the challenge above is to invert the demand system. Under mild assumptions defined in Berry et al. (2013)[3] — which, crucially for our purposes, accommodate both substitutability and complementarity across goods — there exists an inverse σ_j^{-1} such that

$$\delta_{jt} = \sigma_j^{-1} \left(\mathbf{q}_t, \mathbf{p}_t, \mathbf{x}_t^{(2)} \right). \tag{4}$$

The right-hand-side of (4) is only a function of observables and the index on the left-hand-side only contains ξ_{jt}, not the entire J-vector $\boldsymbol{\xi}_t$. Thus, we can now express demand as a system of equations with only one structural error per equation:

$$\beta_j x_{jt}^{(1)} = \sigma_j^{-1} \left(\mathbf{q}_t, \mathbf{p}_t, \mathbf{x}_t^{(2)} \right) - \xi_{jt}, \tag{5}$$

which more closely resembles the structure of typical regression models.

The final challenge is to address endogeneity in the inverse demand equations above. The right-hand-side of (5) not only contains prices, which are subject to the usual endogeneity concerns, but also quantities themselves. Identification will therefore require $2J$ instruments: one J-vector for prices and another J-vector for quantities. Because of the index restriction in (2), the exogeneous characteristics of other products $\mathbf{x}_{-jt}^{(1)}$ are omitted from (5) and can be used to shift quantities independent of ξ_{jt}. Taken together with a set of price instruments $\mathbf{z}_t$ excluded from the demand system such as cost shifters or Hausman instruments, equation (5) can be used to form the moment

[3] The assumption labeled "connected substitutes" in Berry and Haile (2014) requires that there exist a transformation of the demand system exhibiting some degree of strict substitution. As shown in Example 1 of Berry et al. (2013), this condition can be satisfied by a demand system in which all goods are complementary with each other.

conditions:

$$\mathbb{E}\left(\xi_{jt}|\mathbf{z}_t, \mathbf{x}_t\right) = \mathbb{E}\left[\sigma_j^{-1}\left(\mathbf{q}_t, \mathbf{p}_t, \mathbf{x}_t^{(2)}\right) - x_{jt}^{(1)}\middle|\mathbf{z}_t, \mathbf{x}_t\right] = 0, \tag{6}$$

where β_j is normalized to 1 as it cannot be separately identified from the scale of ξ_{jt}. As shown in Berry and Haile (2014), the functions σ_j^{-1} — and thus the demand system σ — are then identified under a completeness condition, the nonparametric equivalent of a standard IV rank condition.

3 Estimation

3.1 A Nonparametric Approach

The moment conditions in (6) lead to a natural flexible GMM estimation approach. Specifically, we follow Compiani (2021) and estimate the inverse demand functions σ_j^{-1} nonparametrically using a sieve estimator with Bernstein polynomials. The choice of Bernstein polynomials is motivated by the fact that they are amenable to a number of economic constraints, such as monotonicity and exchangeability, which also help improve the statistical performance of the estimator. We discuss these constraints in more detail in the next subsection. First, we provide a brief primer on Bernstein polynomial sieve estimators and their approximation properties.

Sieve estimators remain popular tools for nonparametric regression analysis (see reviews in Chen, 2007 or Chen and Qiu, 2013). Here, we focus on sieve estimators constructed with Bernstein basis polynomials, which are defined as:

$$b_{k,m}(u) = \binom{m}{k} u^k (1-u)^{m-k}, \quad k = 0, 1, \ldots, m \tag{7}$$

where m is the degree of the polynomial and $u \in [0, 1]$. A univariate target function can then be approximated with a linear combination of Bernstein basis functions:

$$\sum_{k=0}^{m} \theta_k b_{k,m}(u). \tag{8}$$

In our case, the target estimand is a multivariate function. Let $\mathbf{u} \in [0, 1]^D$ denote the input vector (normalized to the unit hyper-cube) for the target inverse demand function σ_j^{-1}. We can then approximate σ_j^{-1} with the tensor product of Bernstein basis functions:

$$\hat{\sigma}_j^{-1}(\mathbf{u}) = \sum_{k_1=0}^{m} \cdots \sum_{k_D=0}^{m} \theta_{k_1,\ldots,k_D} b_{k_1,m}(u_1) \cdots b_{k_D,m}(u_D). \tag{9}$$

The Bernstein coefficients θ will be estimated by minimizing a GMM objective function based on the moments in (6).

The estimator in (9) is attractive for two reasons. The first is that Bernstein polynomials are sufficiently flexible and can approximate well any continuous function when the coefficients are

suitably chosen. More formally, the following result holds (see, e.g., Gal, 2008).

Result 1. *Let f be a continuous real-valued function on $[0,1]^D$ and define*

$$B_m(f)(\mathbf{u}) = \sum_{k_1=0}^{m} \cdots \sum_{k_D=0}^{m} f\left(\frac{k_1}{m}, \ldots, \frac{k_D}{m}\right) b_{k_1,m}(u_1) \cdots b_{k_D,m}(u_D).$$

Then,

$$\sup_{\mathbf{u}\in[0,1]^D} \left| B_m(f)(\mathbf{u}) - f(\mathbf{u}) \right| \to 0 \quad \textit{as} \quad m \to \infty.$$

In words, when the coefficients on the Bernstein basis functions are equal to the value of the target function f at an equally-spaced grid of points $(\frac{k_1}{m}, \ldots, \frac{k_D}{m})$ for $k_i = 0, 1, \ldots, m$ and $i = 1, \ldots, D$, then the resulting approximation is uniformly good to an arbitrary precision as the degree m grows to infinity.

A second attractive property of the estimator in (9), which also follows from Result 1, is that economic restrictions on $\hat{\sigma}$ can be easily enforced via constraints on the Bernstein coefficients. To illustrate, consider a simple example where the target function $f(u)$ takes one argument and the degree m is set to 2. In this case, the Bernstein polynomial is the sum of three Bernstein basis polynomials:

$$\theta_0(1-u)^2 + \theta_1 2u(1-u) + \theta_2 u^2. \tag{10}$$

Using Result 1, we obtain that if the target function f is monotonically increasing, then the coefficients on the three polynomials must also be increasing, i.e., $\theta_0 \leq \theta_1 \leq \theta_2$.[4] By the same logic, any restrictions on the target function can be translated into constraints on the Bernstein coefficients.

In our case, a complication is that the function being estimated is σ^{-1}, but economic theory yields restrictions on the demand system σ. For this we can use Lemma 1 in Compiani (2021) to translate several constraints on σ into restrictions on σ^{-1}. Moreover, the fact that many of these constraints are linear in the Bernstein coefficients makes them easy to enforce computationally. This is because the GMM objective function is quadratic in the unknown functions σ_j^{-1} and the Bernstein coefficients enter the approximation linearly. Thus, we obtain a well-behaved problem with a convex objective function subject to linear constraints, which can be solved using off-the-shelf global solvers. By the same argument, convex constraints in the Bernstein coefficients can also be accommodated. In contrast, existing estimation procedure such as BLP typically feature non-convex programs, for which solvers that are guaranteed to converge to the global minimum are not available.[5]

[4] More precisely, these conditions become necessary in the limit as $m \to \infty$. Letting the degree of the approximation grow to infinity with the sample size is standard in nonparametric estimation.

[5] Conlon and Gortmaker (2020) find that their implementation of the BLP estimator performs well when optimal instruments are used and restrictions from the supply side are imposed.

3.2 Constraints

The demand specification (i.e., the estimand) outlined in Section 2 is flexible in that no restrictions are placed on the cross-price derivatives of the demand functions σ_j and so the system can admit both substitutes and complements (i.e., positive and negative cross-price effects, respectively). This is in contrast to many widely used market-level demand systems such as BLP which are based on assumptions of discrete choice and additively separable utility, thus ruling out complementarity. The estimator in Section 3 is also flexible in that it comes with uniform approximation guarantees. Although there are many other demand systems based on flexible functional forms[6] (e.g., the almost ideal demand system of Deaton and Muellbauer, 1980), these models are typically made stochastic by appending an iid mean zero error term to the end of each estimating equation and therefore do not incorporate "structural" errors. That is, the errors cannot be interpreted as unobserved product-specific demand shifters — giving rise to the endogeneity of prices — that jointly enter the demand equation for each good (Berry and Haile, 2021).

Flexibility does come at a cost, however. Flexible estimators necessarily contain many parameters and can therefore suffer from a curse of dimensionality. In our case, the number of Bernstein coefficients used in the approximation of σ_j^{-1} grows exponentially in the number of goods, and so a fully unrestricted estimator will be practically intractable for a moderate to large number of goods. A fully unrestricted estimator is also undesirable in that the resulting system is not guaranteed to be consistent with properties of a valid Marshallian demand system. As discussed in Compiani (2021), one of the main reasons that Bernstein polynomials are attractive is that they are amenable to a variety of microfounded constraints. In what follows, we briefly discuss different constraints that can be imposed on the nonparametric estimator in order reduce to the number of parameters and/or ensure that various restrictions required by economic theory are satisfied. Note that not all of the following constraints must be imposed simultaneously — for example, in the empirical application we do not impose a lack of income effects. The decision of which constraints to include will likely be governed by the empirical context. A summary of restrictions is provided in Table 1.

Exchangeability A demand system is exchangeable if σ_j is the same for all j and it is invariant to permutations of product characteristics $(\delta_k, p_k, x_k^{(2)})$ for all other goods $k \neq j$. Exchangeability is satisfied in mixed logit models under the standard assumption that the distribution of random coefficients is the same across all goods. Moreover, systematic differences across goods can be accounted for via product fixed effects in the indices δ_{jt}. As shown in Compiani (2021), the approximation properties of Bernstein polynomials (Result 1) can be used to translate exchangeability into equality constraints on the Bernstein coefficients. Thus, this constraint effectively reduces the number of parameters that need to be estimated, alleviating the curse of dimensionality.

[6] The standard definition of a "flexible functional form" is that it can provide a second-order approximation to an arbitrary twice differentiable function. While flexible functional forms are often used to approximate utility or cost functions (e.g., Deaton and Muellbauer, 1980), it is also equivalent to target first-order approximations to the demand system directly (Pollak and Wales, 1992, ch. 3). For a system of J goods, this requires at least $1 + J + J(J+1)/2$ free parameters in each demand equation. Our nonparametric estimator satisfies this condition.

Table 1: Summary of Constraints

	Economic Meaning
1. Exchangeability	Demand σ_j invariant to permutations of $(\boldsymbol{\delta}_{-j}, \mathbf{p}_{-j}, \mathbf{x}_{-j}^{(2)})$
2. Price inside the index	Constant marginal rate of substitution between price and other characteristics in the index
3. Symmetry of $\mathbb{J}_\sigma^p(\delta, p)$	No income effects
4. Symmetry of $\mathbb{J}_\sigma^\delta(\delta, p)$	Utility of good j linear in δ_j and doesn't depend on δ_k for $k \neq j$
5. Diagonal dominance of $\mathbb{J}_\sigma^\delta(\delta, p)$	Own effects dominate cross effects
6. Own-good monotonicity	Demand σ_j increases in $(\delta_j, -p_j)$

	Necessary for invertibility?	Necessary for identification?	Reduction in # parameters?	Constraint
1. Exchangeability	No	No	Yes	Linear
2. Price inside the index	No	No	Yes	Linear
3. Symmetry of $\mathbb{J}_\sigma^p(\delta, p)$	No	No	No	Nonlinear*
4. Symmetry of $\mathbb{J}_\sigma^\delta(\delta, p)$	No	No	No	Linear
5. Diagonal dominance of $\mathbb{J}_\sigma^\delta(\delta, p)$	No	No	No	Linear
6. Own-good monotonicity	Yes	Yes	No	Linear

*The symmetry of $\mathbb{J}_\sigma^p(\delta, p)$ constraint is nonlinear only if price is not inside the index; otherwise, it is linear.

Index Restriction Instead of allowing prices p_{jt} and covariates $\mathbf{x}_t^{(2)}$ to enter the demand system fully flexibly, one could include them in the linear indices:

$$\delta_{jt} = \beta_{1j} x_{jt}^{(1)} + \mathbf{x}_{jt}^{(2)\prime} \beta_{2j} - \alpha_j p_{jt} + \xi_{jt}. \tag{11}$$

This leads to simpler demand equations $s_{jt} = \sigma_j(\boldsymbol{\delta}_t)$, which can again be inverted as $\delta_{jt} = \sigma_j^{-1}(\mathbf{q}_t)$. The advantage is that the estimands σ_j^{-1} are now a function of J arguments only, instead of $2J + n_{x^{(2)}}$, where $n_{x^{(2)}}$ denotes the dimension of the covariates $\mathbf{x}^{(2)}$. Estimating functions of fewer arguments reduces the number of parameters and thus alleviates the curse of dimensionality. Note that including p_{jt} in the linear index δ_{jt} does not imply that *demand* is linear in prices, but instead simply requires that the marginal rate of substitution between price and other characteristics in the index be constant. In BLP-type models, for example, this index restriction is consistent with price entering the indirect utility function with a non-random coefficient.[7] This highlights a trade-off between the amount of heterogeneity allowed in the model and the number of parameters that need to be estimated.

Slutsky Symmetry Slutsky symmetry refers to the symmetry of the Slutsky substitution matrix and is required for integrability. Since we observe and model uncompensated (Marshallian) demand, however, we can only impose Slutsky symmetry if income effects are zero. When prices enter the

[7] Because preference heterogeneity is only weakly identified with aggregate data (Bodapati and Gupta, 2004; Albuquerque and Bronnenberg, 2009; Dunker et al., 2017), it is common practice to only place random coefficients on a subset of variables in BLP models.

linear indices δ_{jt} (see "Index Restriction" above), symmetry can be imposed via linear constraints on the Bernstein polynomial, whereas the constraints become nonlinear — and generally non-convex — when prices are not part of the index (Compiani, 2021).

Diagonal Dominance Diagonal dominance requires the magnitude of the own-price effects to be at least as large as the sum of magnitudes of the cross-price effects. Specifically, the demand system σ satisfies (column) diagonal dominance if:

$$\sum_{k \neq j} \left| \frac{\partial \sigma_k}{\partial p_j} \right| \leq \left| \frac{\partial \sigma_j}{\partial p_j} \right|. \tag{12}$$

This restriction is satisfied by many common demand models, including BLP. Compiani (2021) shows that diagonal dominance can be enforced via linear inequality constraints on the Bernstein coefficients. Although not identical, a similar version of this property appears as a consequence of the homogeneity and adding-up (Cournot aggregation) restrictions required for integrability.[8] The only difference is that homogeneity and adding-up require that the *price-weighted* own-effect dominate the sum of *price-weighted* cross-effects. Also note that Anderson and Vilcassim (2001) propose a condition that amounts to diagonal dominance of the *profit function* with respect to prices in order to ensure finite optimal prices. The intuition is that the retailer's first-order conditions depend directly on derivatives of the aggregate demand function, and so optimal prices can approach infinity when cross-product substitution is left "unbounded."

Own-Good Monotonicity Monotonicity requires the demand for each product to be increasing in its own quality index δ_{jt} and decreasing in its own price. Requiring negative own-price effects with respect to the uncompensated (Marshallian) demand function will, in turn, ensure negativity of the Slutsky substitution matrix which is required for integrability (assuming nonnegative income effects). Again, by the approximation properties in Result 1, monotonicity can be imposed easily through inequality constraints on the Bernstein coefficients.

3.3 Sources of Misspecification

While the constraints outlined above are useful for ensuring the validity or tractability of the demand system, standard parametric models impose additional restrictions that could lead to misspecification in estimation. Here we focus on two specific sources. The first is a misspecification of the functional form of the demand system. The second is a misspecification of the arguments of the demand system that arises by omitting relevant goods. Both sources are consequences of decisions that must be made when estimating demand, and are therefore relevant for most empirical work.

[8] See the discussion of Propositions 2.E.1 and 2.E.2 in Mas-Colell et al. (1995).

Imposing overly restrictive functional forms

Many common but tightly parameterized specifications of demand are nested within the general model outlined in Section 2. Here we focus on logit and mixed logit models. To make connections to the nonparametric model above, we will show how the estimating equations in (5) can be expressed for each parametric model. First, consider the logit model with utilities $u_{ijt} = \beta_j x_{jt}^{(1)} - \alpha p_{jt} + \xi_{jt} + \varepsilon_{ijt}$. Berry (1994) shows that the logit demand system can be inverted as

$$\beta_j x_{jt}^{(1)} = \left[\log q_{jt} - \log q_{0t}\right] + \alpha p_{jt} - \xi_{jt}, \tag{13}$$

where now q_{jt} denotes the market share of good j. This is a special case of equation (5) and therefore the logit model is subsumed by the model in Section 2. In the case of the mixed logit model, the inverse demand system does not have a closed form solution, but it is still well-defined. In particular, we have

$$\beta_j x_{jt}^{(1)} = \tilde{\delta}_j \left(\mathbf{q}_t, \mathbf{p}_t, \mathbf{x}_t^{(2)}; \Sigma\right) - \xi_{jt}. \tag{14}$$

where $\tilde{\delta}_j(\cdot)$ are the mean utilities — characterized by nonlinear parameters Σ — that can be solved for via the BLP contraction mapping (Berry et al., 1995). Again, this shows that any mixed logit model with a non-random coefficient on $x_j^{(1)}$ is subsumed by the model in (5).

These standard models may be misspecified in at least two ways. First, the discrete choice assumption underlying (13) and (14) restricts the sign of all cross-price effects to be positive and so the model will clearly be misspecified if the true system includes complementary goods. Second, even if all goods are imperfect substitutes, the logit functional form can lead to biased own- and cross-price effects. In contrast, our estimator allows for both complementarity and substitution and does not impose the logit functional form. We will also consider an intermediate specification: a nonparametric model that restricts all cross-price effects to be positive. Like standard discrete choice models, this imposes strict substitution among all goods. On the other hand, it avoids placing a parametric functional form on the demand system and is in this sense more flexible than logit-based models. In our empirical analysis, we will compare these models with a focus on assessing the costs of functional form misspecification in the retailer pricing problem.

Omitting relevant goods

Misspecification of demand can also arise in the decision of *which* goods to include in the analysis. The specific concern is that the prices of related goods, when omitted from a demand system, may act as unobserved confounders and therefore bias estimates of key economic objects such as own-price elasticities. In this section, we investigate the nature of this bias using the underlying model structure alone. In other words, based on the functional form of demand, what can be said about the bias in price effect estimates when related goods are omitted?

If demand was specified with a simpler regression-based functional form, then the sign and magnitude of this bias would follow from standard omitted variable bias (OVB) results. For example,

in a linear demand model:

$$q = \alpha p + \tilde{\mathbf{p}}'\beta + \epsilon \tag{15}$$

the OVB in the ordinary least squares estimator of the own-price effect when omitting $\tilde{\mathbf{p}}$ is:

$$\text{Bias}(\hat{\alpha}_{\text{OLS}}) = \frac{\text{Cov}(p, \tilde{\mathbf{p}})}{\text{Var}(p)}\beta. \tag{16}$$

The sign of the bias is thus determined by sign of the coefficient on the omitted variable(s) and the sign of the covariance between the included price and omitted price(s). Having this type of expression for the bias can be useful for guiding decisions about consumer choice sets and market boundaries. In particular, (16) suggests that we can mitigate OVB in a linear demand model by including all goods that are related to the focal good ($\beta \neq 0$) and whose prices are related to the focal good's price ($\text{Cov}(p, \tilde{\mathbf{p}}) \neq 0$).

The challenge is that the nonparametric demand system in Section 2 is not a linear regression model and is not estimated by ordinary least squares. Instead, our nonparametric demand system is highly nonlinear in prices and error terms, and σ^{-1} (not σ) is estimated via nonparametric IV regression. So to what extent do the standard OVB results apply? Chalak (2019) extends the OVB result above in a few key ways that relate to our nonparametric analysis. Specifically, he provides OVB results for linear models with an instrument for the included price p, denoted by z, as well as more general nonseparable models $q = \sigma(p, \tilde{\mathbf{p}}, \epsilon)$ both with and without instruments.[9] For example, in a linear model with an instrument for p, the bias becomes

$$\text{Bias}(\hat{\alpha}_{\text{IV}}) = \frac{\text{Cov}(z, \tilde{\mathbf{p}})}{\text{Cov}(z, p)}\beta \tag{17}$$

which is similar to (16) except that now we must check whether the included *instrument* (not price) is correlated with omitted prices. The functional form for the nonlinear, nonseparable case is more complicated (see theorems 5.1(ii) and 6.1(ii) in Chalak, 2019), but the same intuition holds: bias is governed by the sign of the omitted cross effect(s) and the relationship between the included price/instrument and omitted price(s). Although deriving the exact OVB formula for nonparametric demand models is beyond the scope of this paper, we use the intuition above to guide our empirical analysis of bias arising in smaller, misspecified demand systems.

[9] One source of endogeneity bias is the presence of unobserved confounders. In principle, a valid instrument for the included price p can produce consistent estimates of the price coefficient α even in the presence of omitted prices $\tilde{\mathbf{p}}$. However, omitting prices of relevant goods poses a threat to the validity of the instrument. In a smaller, misspecified model, the instrument satisfies exogeneity only if it is uncorrelated with both the "true" error term from the correctly specified model and the omitted price(s). The latter condition is not present when all relevant prices are included in the model, and will typically be hard to satisfy with the usual set of instruments used in demand estimation (e.g., cost shifters, BLP instruments, Hausman instruments).

4 Monte Carlo Simulations

In this section, we illustrate the flexibility of our approach by showing that the nonparametric estimator is able to capture complex patterns of substitution/complementarity. We also investigate the effects of misspecified demand boundaries on estimated elasticities. Consider a setting in which the first $J-1$ goods are substitutes to each other, but are complementary to the Jth good. We generate prices as $p_{jt} = \frac{e^{\tilde{p}_{jt}}}{1+e^{\tilde{p}_{jt}}}$, where $\tilde{p}_{jt}$ are iid standard normal across products j and across markets t. Further, we let x_{jt} be uniformly distributed on the unit interval (also iid across products and markets), and let the unobservables ξ_{jt} be iid normal with mean zero and standard deviation 0.15. Then, letting $\delta_{jt} = -2p_{jt} + x_{jt} + \xi_{jt}$, we define the quantity of good j as:

$$q_{jt} = \exp\left(\delta_{jt} - 0.15 \sum_{k<J,\ k\neq j} \delta_{kt} + 0.15\delta_{Jt}\right) \tag{18}$$

for goods $j = 1, \ldots, J-1$ and

$$q_{Jt} = \exp\left(\delta_{Jt} + 0.15 \sum_{k<J} \delta_{kt}\right) \tag{19}$$

for the final Jth good. Note that the sign of the coefficients on the indices δ_{kt} $(k < J)$ and δ_{Jt} produce the desired block-wise pattern of substitutes and complements. The elasticities induced by this system will be:

$$\mathcal{E}_{jkt} = \frac{\partial q_{jt}/q_{jt}}{\partial p_{kt}/p_{kt}} = \begin{cases} -2p_{kt} & \text{if } j = k \\ -0.15 \cdot 2p_{kt} & \text{if } j = J \text{ or } k = J, \text{ for } j \neq k \\ 0.15 \cdot 2p_{kt} & \text{otherwise.} \end{cases}$$

We generate 100 data sets from the model above, with each having $T = 10,000$ markets. We then apply the nonparametric estimator to each data set where we impose the following constraints: an index restriction (with price inside the index), diagonal dominance, weak substitutability and exchangeability among the first $J-1$ goods, and own-good monotonicity for all goods. The relationship between the first $J-1$ goods and the Jth good is left unrestricted.

Elasticity estimates from a $J = 3$ good system are reported in column (I) of Table 2. We specifically report estimates of the sampling distribution of the median. That is, for each data set we compute the median elasticity across observations, and then report the mean and standard deviation of those point estimates. Because we generate prices that are defined on the unit interval and centered at 0.5, the true own elasticities are equal to $\mathcal{E}_{jj} = -1$, the true cross elasticities between goods $j < J$ and $k < J$ are equal to $\mathcal{E}_{jk} = 0.15$, and the true cross elasticities between goods $j < J$ and $k = J$ are equal to $\mathcal{E}_{jk} = -0.15$. We find that the nonparametric estimator accurately recovers the true values of all model parameters. Elasticity estimates from larger $J = 4$ and $J = 5$ good systems, which also accurately recover true values, are reported in Appendix A.

Table 2: Elasticity Estimates

	True Value	(I) $J=3$	(II) $J=3$ Subst	(III) $J=2$ $\rho=0$	(IV) $J=2$ $\rho=0.7$	(V) $J=2$ $\rho=-0.7$
$\mathcal{E}_{11}$	-1.00	-0.98 (0.026)	-1.01 (0.026)	-0.98 (0.026)	-1.07 (0.027)	-0.89 (0.023)
$\mathcal{E}_{22}$	-1.00	-0.98 (0.024)	-1.00 (0.023)	-0.98 (0.023)	-1.06 (0.025)	-0.88 (0.022)
$\mathcal{E}_{33}$	-1.00	-1.02 (0.033)	-1.05 (0.033)			
$\mathcal{E}_{12}$	0.15	0.14 (0.005)	0.11 (0.006)	0.14 (0.005)	0.04 (0.004)	0.24 (0.007)
$\mathcal{E}_{21}$	0.15	0.14 (0.005)	0.11 (0.005)	0.14 (0.005)	0.04 (0.004)	0.24 (0.007)
$\mathcal{E}_{13}$	-0.15	-0.15 (0.006)	0.00 (0.000)			
$\mathcal{E}_{31}$	-0.15	-0.15 (0.005)	0.00 (0.000)			
$\mathcal{E}_{23}$	-0.15	-0.14 (0.005)	0.00 (0.000)			
$\mathcal{E}_{32}$	-0.15	-0.15 (0.006)	0.00 (0.000)			

Notes: In column (I), we impose substitution between the first two goods, but leave the $\{1,2\}$-$\{3\}$ effects unrestricted. In column (II), we erroneously impose substitution between all three goods. In columns (III)-(V), we fit misspecified models where the third good is omitted and under varying degrees of price correlation ρ. Standard errors are reported in parentheses.

So far, we have shown that the nonparametric estimator is flexible and can accurately recover elasticities in a demand system with a mix of substitutes and complements. Our next objective is to highlight the possible effects of misspecified boundaries in the estimation problem, as discussed in Section 3.3. We first estimate a model that erroneously assumes weak substitution between all three goods. The estimates from this misspecified model are reported in column (II). Although the own elasticities remain accurately recovered, we find a clear bias in all cross elasticities. Specifically, the cross elasticities involving the third good are all estimated to be very close to zero because of the binding positivity constraint which, in turn, leads to a downward bias on the estimated elasticities between goods 1 and 2.[10]

Next, we estimate a misspecified model that excludes the third good from the demand system entirely. We also relax the assumption that prices are iid across goods, and define $\rho = \text{Corr}(p_{jt}, p_{kt})$ to be the correlation between the price of good $j < J$ and omitted good $k = J$. We then consider three scenarios: one where $\rho = 0$ and prices remain iid across goods, a second where $\rho = 0.7$ and thus the price of the third good is positively correlated with the price of the first two goods, and a third where $\rho = -0.7$. When prices are correlated and the scope of the demand system is misspecified, a standard omitted variable bias problem arises.

Columns (III)-(V) report estimated elasticities from this smaller, misspecified demand system with $J = 2$ goods. In column (III), we find that both the own and cross elasticities can still be accurately recovered when there is no underlying correlation in prices. This suggests that, conditional on having a sufficiently flexible estimator of substitution patterns, the omission of related goods alone is not enough to generate biases in elasticity estimates. However, in columns (IV) and (V) where $\rho \in \{0.7, -0.7\}$, we find clear biases in both own and cross elasticity estimates. Moreover, the sign of this bias is governed by sign of the product of ρ and the omitted cross

[10] Compiani (2021) shows that, under a different DGP with complementarity, fitting a discrete choice model which assumes substitution across goods yields substantial bias in both the own- and the cross-price elasticities.

elasticity, which is consistent with the omitted variable bias intuition discussed in Section 3.3. For example, we find that the estimated elasticities are biased *downwards* in column (III) where $\rho \times \mathcal{E}_{j3} = 0.7 \times -0.15 < 0$ and biased *upwards* in column (IV) where $\rho \times \mathcal{E}_{j3} = -0.7 \times -0.15 > 0$.

5 Data

Our empirical analysis uses data from one major grocery retail chain in the United States.[11] The retailer has nearly 500 stores spanning five states. Our sample consists of weekly UPC-level quantities, prices, feature promotion activity, wholesale prices, and marginal costs[12] across all departments in each store during the years 2014-2016. Ideally, we would estimate one large demand system that included products from most grocery departments. Such a system would allow us to measure all cross-category substitution and then, in some sense, define market boundaries solely from the data. However, the size of such a system would make our nonparametric approach — as well as most all structural approaches — intractable. We must therefore trade-off scope and flexibility with tractability. We do this by aggregating products to the category level and estimating a multi-category system separately for different "product groups." For example, Baking Goods is a product group with two categories (cake mix and frosting) and Refrigerated Juice is a product group with five categories (orange juice, lemonade, fruit juice, other juice, and iced tea). This unit of analysis makes it feasible to flexibly estimate cross-category substitution, while still allowing for a wider set of categories than typically used in market-level demand estimation.

Defining products at the category level requires us to aggregate UPC-level quantities and prices. We define quantities as the total volume (in ounces) across all UPCs in the category and define prices as revenue-weighted prices, where weights are computed for each category-store-year. We also use the same revenue weights to construct category-level promotion, wholesale price, and marginal cost variables. Estimating market demand also raises questions of price endogeneity, which we address using a battery of fixed effects and instruments. We construct Hausman IVs — i.e., the average price of the same product in other markets (Hausman, 1996) — to instrument for price.

We construct nine different data sets from product groups that vary in the nature of product differentiation and expected substitution. A list of these nine groups and all associated categories can be found in Table 3. Groups like Butter/Margarine/Spreads, Crackers, or Hot Dogs all consist of categories that are likely substitutes. Other groups like Baking Goods, Detergent/Softener, and Jams/Jellies/Peanut Butter consist of categories that are likely complements.[13] We also include groups like Refrigerated Juice where the nature of cross-category substitution is less clear a priori. Substitution in a group like cereal, one of the classic markets for applying logit models (Nevo, 2001), is also perhaps not obvious. Households with kids, for example, may purchase multiple varieties across the adult and kids categories and so assumptions of discrete choice may be violated. Because

[11] The data are provided by DecaData (http://decadata.io).

[12] Marginal cost is defined as wholesale price less any trade deals between the manufacturer and retailer.

[13] Some groups actually consist of both complements and substitutes. For example, regular and organic peanut butter are likely substitutes with each other, but complement regular and organic jams/jellies.

Table 3: Product Category Descriptions

Product Group	Categories
Baking Goods	(2) Cake Mix, Frosting
Butter/Margarine/Spreads	(3) Butter, Margarine, Spreadable Butter
Canned Fish	(5) Tuna (Chunk-Light, Chunk-White, Solid-White), Sardines, Salmon
Cereal	(4) Adult, All Family, Kids, Better For You
Crackers	(3) Savory, Spray Butter, Saltines
Detergent/Softeners	(4) Laundry Detergent (Dry, Liquid), Fabric Softener (Dry, Liquid)
Hot Dogs	(3) Beef, Meat, Turkey
Jams/Jellies/Peanut Butter	(4) Peanut Butter (Regular, Organic), Jams/Jellies (Regular, Organic)
Refrigerated Juice	(5) Orange Juice, Lemonade, Fruit Juice, Other Fruit Juice, Iced Tea

we flexibly estimate the demand function directly, we relax standard restrictions on utility/demand and can thus let the data tell us about the nature of substitution.

6 Descriptive Evidence Across Many CPG Groups

We begin our empirical analysis with a set of descriptive results characterizing demand and supply-side boundaries. On the demand side, our goal is to demonstrate that appreciable cross-category effects exist in our data, which then motivates the need for a flexible structural model of cross-category substitution that can be used to carry out pricing counterfactuals and quantify costs of model misspecification. To this end, we estimate a large number of log-log demand models across all nine product groups listed in Table 3. The product is defined as a category and q_{ist} represents quantity sales for category i at store s in week t. Also, let $\mathcal{G}_i$ denote the product group associated with category i. For example, if i is cake mix then $\mathcal{G}_i = \{\text{cake mix, frosting}\}$ is the set of categories in Baking Goods. We follow Hitsch et al. (2021) and specify demand for good i as a function of prices and promotions for all goods within the same product group.

$$\log q_{ist} = \alpha_i + \sum_{j \in \mathcal{G}_i} \beta_{ij} \log p_{jst} + \sum_{j \in \mathcal{G}_i} \theta_{ij} d_{jst} + \phi_i(t) + \varepsilon_{ist} \tag{20}$$

Here, $\log p_{jst}$ is the log price of category j, d_{jst} is the volume of feature advertising for category j, and $\phi_i(t)$ are time fixed effects (season and holiday dummmy variables). We estimate the model above separately for each category-ZIP code, thus pooling information across all stores within a ZIP code. In Table 4 we report summary statistics of the own-price and cross-price elasticities for each product group. We specifically report the mean, standard deviation, and share of positive (or negative) elasticity estimates for estimates that are significant at the 5% level. The complete distributions of own-price and cross-price elasticities, as well as the own-promotional effects are reported in Appendix B.

We find substantial heterogeneity in the sign and magnitude of cross-price elasticities. For six of the nine product groups, the average cross-category elasticity is positive. The largest effects come

Table 4: Summary of Elasticity Estimates from the Log-Log Model

	Cross-Price Elasticity				Own-Price Elasticity			
Product Group	Mean	SD	% Pos.	% Sig.	Mean	SD	% Neg.	% Sig.
Baking Goods	-0.99	0.88	11.74	36.16	-1.41	0.53	98.71	53.01
Butter/Margarine/Spreads	1.23	1.66	77.14	19.18	-1.44	1.59	79.81	78.72
Canned Fish	0.26	1.13	64.25	10.96	-0.75	0.72	91.53	44.66
Cereal	0.61	2.00	69.00	18.93	-2.27	1.16	97.02	55.14
Crackers	1.15	0.76	94.39	22.79	-2.00	0.74	99.56	82.92
Detergent/Softener	-0.48	0.85	25.28	22.03	-0.89	0.75	97.72	41.99
Hotdogs	0.29	1.30	64.38	30.00	-1.34	0.82	93.73	55.34
Jams/Jellies/Peanut Butter	-0.60	1.86	34.74	19.98	-1.81	0.81	98.30	60.27
Refrigerated Juice	0.23	2.36	52.95	25.08	-2.03	1.47	92.70	75.71

Notes: Log-log models are estimated at the category-ZIP code level. The mean, standard deviation, and share of positive (or negative) price elasticities are conditional on estimates significant at the 5% level.

from the Butter/Margarine/Spreads and Crackers product groups, with average cross-elasticities of 1.23 and 1.15 respectively. Among these six groups, we still find substantial variation in the estimated cross elasticities. For example, in Refrigerated Juice the average cross elasticity is 0.23 but roughly 47% of the precisely estimated elasticities are negative. While some of the heterogeneity in these estimates can be attributed to lack of economic structure in a log-log demand system, the large mass of negatively estimated elasticities that is persistent across many groups suggest that substitution across categories may be more nuanced. The remaining three product groups (Baking Goods, Detergent/Softener, and Jams/Jellies/Peanut Butter) exhibit negative cross elasticities on average, suggesting strong complementarity across categories. This is consistent with existing work that has documented complementarities between cake mix and frosting (e.g., Manchanda et al., 1999; Ma et al., 2012), and laundry detergent and fabric softener (e.g., Song and Chintagunta, 2006; Mehta, 2007; Song and Chintagunta, 2007).

The estimated own-price elasticities are all negative, on average, for each product group. We find that demand is most elastic in the Cereal and Refrigerated Juice groups, where the average own elasticities are -2.27 and -2.03, respectively. Although the share of negative own elasticities is quite high across groups, we consistently find a small fraction of positive (i.e., "incorrectly signed") and statistically significant estimates. Again, this is partly due to the lack of economic structure in the log-log system and is fairly typical when applying estimating regression models on store-level scanner data (Blattberg and George, 1991; Boatwright et al., 1999; Hitsch et al., 2021). In these regressions, we are also ignoring the potential endogeneity of prices, which would generate an upward bias in the estimated own elasticities. We address both of these limitations in our more structural, nonparametric analysis that follows, where we impose monotonicity restrictions to ensure negative own-price effects and also instrument for prices to alleviate endogeneity concerns.[14]

[14] One additional concern when estimating demand in CPG markets is dynamics and the storable goods demand problem, which can also bias estimates of price elasticities (Erdem et al., 2003; Sun et al., 2003). In Appendix C, we implement the descriptive tests of Hendel and Nevo (2003) and find minimal evidence of stockpiling across all groups.

Table 5: Correlations between Prices and Instruments

	Corr(p_i, p_j)			Corr(z_i, p_j)			Corr(z_i, p_i)		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
Baking Goods	0.52	0.52	0.52	0.54	0.59	0.63	0.84	0.89	0.93
Butter/Margarine/Spreads	0.02	0.09	0.16	-0.01	0.07	0.20	0.43	0.73	0.99
Canned Fish	-0.09	0.14	0.65	-0.64	0.02	0.82	0.21	0.44	0.57
Cereal	0.12	0.28	0.70	0.13	0.39	0.76	0.49	0.75	0.97
Crackers	0.13	0.27	0.56	0.11	0.28	0.63	0.83	0.88	0.96
Detergent/Softener	0.01	0.09	0.35	0.02	0.18	0.59	0.13	0.55	0.73
Hotdogs	0.15	0.15	0.16	0.09	0.13	0.21	0.68	0.76	0.84
Jams/Jellies/Peanut Butter	0.10	0.27	0.57	0.07	0.28	0.74	0.55	0.74	0.90
Refrigerated Juice	-0.32	-0.03	0.13	-0.58	-0.04	0.22	0.62	0.77	0.89

Notes: Correlations between two prices (p_i, p_j) and between instrument-price pairs (z_i, p_j) are estimated across all categories within each product group. The minimum, mean, and maximum correlation values are reported for each group.

We now turn to the supply side, where our goal is to document correlations in prices and instruments across categories. As discussed in Section 3.3, correlations between prices and instruments are relevant in the delineation of market boundaries to the extent that they contribute to an omitted variable bias problem. That is, key demand elasticities may be biased if we omit goods from our estimating equations that are both related to the focal good (i.e., exhibit a non-zero cross-effect) and whose prices are correlated with the focal good's price/instrument.

Table 5 reports three types of correlations for each product group. The first is Corr(p_i, p_j), which is the correlation between prices of categories i and j, with $i \neq j$. The second is Corr(z_i, p_j), which is the correlation between the Hausman instrument for category i and the price of category j, with $i \neq j$. The third is Corr(z_i, p_j) with $i = j$. For each type of correlation, we report the minimum, mean, and maximum values across category pairs. Across all nine product groups, we find moderate correlations between prices (p_i, p_j) — ranging from -0.32 to 0.70 — and between prices and instruments (z_i, p_j) — ranging from -0.64 to 0.82. On average, both types of correlations are away from zero and positive, indicating that omitted variable bias may indeed be an issue if one were to omit relevant goods from the demand system. We also find that Corr(z_i, p_i) tends to be positive and large in magnitude, which is to be expected for Hausman instruments.

Together, the descriptive analysis above highlights a few key results. First, within each product group, there appears to be appreciable substitution across categories. Second, the nature of substitution is nuanced and many product groups contain combinations of strict substitutes and complements. Third, log-log models lack proper economic structure, which is evidenced by the subset of incorrectly signed own elasticities. Fourth, many categories exhibit sizeable correlations in prices which can contribute to an omitted variable bias problem in estimation. We therefore move onto a more structural model of demand which is both flexible and valid, allowing us to properly measure both the extent of cross-category substitution/complementarity as well as the cost of misspecified boundaries of competition.

7 A Structural Demand Analysis for Refrigerated Juice

In this section, we present a more structural analysis of demand using the model and nonparametric estimation approach outlined in Sections 2 and 3. For illustrative purposes, we focus on the Refrigerated Juice product group. There are a few reasons why this group is particularly attractive. First, while the orange juice category has long been used for demand estimation and pricing research (Montgomery, 1997; Chintagunta et al., 2003; Nair et al., 2005; Dubé et al., 2008, 2010), few studies have measured substitution across orange juice and other refrigerated juice categories. Quantifying these cross-category effects can help us determine whether orange juice can be treated as its own market and whether the optimal pricing function for orange juice products should consider the demand for related juices, and vice versa. Second, the nature of substitution across juice categories is complicated. This product group contains many leading CPG manufacturers with brands in multiple categories (e.g., Simply Orange and Simply Lemonade). Thus, the presence of strong brand effects or coordinated marketing efforts could lead to some form of demand spillover across categories (Smith et al., 2019). Moreover, it is not clear whether different flavors of juice should be considered strict substitutes. Our descriptive results in Section 6 showed that cross elasticities are very heterogeneous in Refrigerated Juice, with roughly half of the precisely estimated cross-price elasticities being positive and the other half negative. It is plausible that forms of brand complementarities or demand for variety leads to joint purchases spanning multiple categories. In this case, assumptions of discrete choice underlying workhorse models like BLP become less tenable. For these reasons, we view this product group as a useful laboratory to illustrate our approach.

7.1 Nonparametric Specification and Alternative Models

We estimate a nonparametric demand system for all five categories in the Refrigerated Juice product group. Besides prices, we include the following exogenous demand shifters $\mathbf{x}_t$: feature promotional activity, season, holiday, 5-digit ZIP code and category dummies. We use the Hausman instruments described in Section 5 as excluded instruments for prices. Appendix D reports R^2 and F-test statistics from first-stage regressions of prices on exogenous characteristics and Hausman instruments. The results show that the instruments are relevant. Furthermore, we impose the following four constraints from Section 3.2: (i) price and all $\mathbf{x}_t$ inside the index (i.e., all of our $\mathbf{x}_t$ variables are $\mathbf{x}_t^{(1)}$ in the notation of Section 2, so that there are no $\mathbf{x}_t^{(2)}$ in the model); (ii) exchangeability between the products in the set $\mathcal{J}' = \{\text{Lemonade, Fruit Juice, Other, Iced Tea}\}$, but not across $\{\text{Orange Juice}\}$ and $\mathcal{J}'$;[15] (iii) diagonal dominance; and (iv) negative own-price effects. This leads

[15] As discussed in Compiani (2021), exchangeability is a practically useful assumption that helps reduce the dimension of the estimation problem. We only impose exchangeability among all categories other than orange juice in order to keep the relationship between orange juice and other categories as flexible as possible. This decision is in part motivated by the fact that orange juice is a widely used category in empirical work and so we are especially interested in substitution between orange juice and other flavor categories. Note that the inclusion of category fixed effects means that the model allows for systematic unobserved differences even among the fours categories assumed to be exchangeable.

to the following specification of the product index and demand equations:

$$\delta_{jt} = \mathbf{x}'_{jt}\beta_j - \alpha_j p_{jt} + \xi_{jt} \tag{21}$$

$$q_{jt} = \sigma_j(\boldsymbol{\delta}_t) \tag{22}$$

which we refer to as the "NPD-Flex" model (for Nonparametric Demand Model).

We also estimate two alternative structural models. The first is a nonparametric model with the restrictions stated above as well as a positive sign constraint on all cross-price effects.[16] This additional sign constraint creates a nonparametric demand model for strictly substitutable goods, which we refer to as "NPD-Subst." The second is a BLP model which, by construction, imposes substitution across all goods. Utility is specified as a linear function of price, feature promotions, and product intercepts, and random coefficients are placed on the intercept, category dummies, and price. The model is estimated using the PyBLP package (Conlon and Gortmaker, 2020).[17]

In summary, both NPD models are flexible models in the sense that demand is specified as a nonseparable function of prices, other product characteristics, and structural error terms, and this function is estimated nonparametrically. Further, the NPD-Flex model imposes no constraints on the sign of cross-price effects and so the model can accommodate both substitutable and complementary goods. In contrast, the NPD-Subst and BLP models only admit positive price effects and thus rule out any form of complementarity. BLP imposes further functional form restrictions as discussed in Section 3.3.

7.2 Model Comparison

Given our goal of assessing the costs of misspecification, we need to be able to compute profits based on a ground truth model of demand. Therefore, our first task is to assess the fit of each model to the data. By comparing model fit statistics, we can see if there is empirical support for a model allowing for more flexible demand derivatives and cross-category complements. Each NPD model also contains a tuning parameter m which controls the complexity of the nonparametric estimator. Specifically, each unknown function is estimated using an m-degree Bernstein polynomial in each of its arguments (leading to an mJ-degree polynomial for each function). We also want to use the data to select an optimal value of m for each model.

Our procedure for evaluating model fit and selecting tuning parameters is as follows. We first split our data into two subsamples: a *selection sample* (30% of weeks, randomly selected) and an *estimation sample* (the remaining 70% of weeks). This initial sample-split is done to preserve the validity of any standard errors computed on elasticities.[18] To compute model fit statistics (for model

[16] The constraint is technically a negative sign constraint on the cross-derivative of $\sigma_j(\cdot)$ with respect to the index $\delta_k(p_k, \xi_k)$. Since the index is a linear function of $-p_k$, a negative constraint with respect to the index δ_k is the same as a positive constraint with respect to the price p_k.

[17] We adopt the following "best practices" suggested in Conlon and Gortmaker (2020): (i) the fixed point problem is solved via the SQUAREM algorithm; (ii) the GMM objective function is optimized using SciPy's L-BFGS-B algorithm with a tighter tolerance of 1e-14; (iii) integrals are approximated via the quasi Monte Carlo method with scrambled Halton draws. We also use the same Hausman IVs as in both NPD specifications to instrument for price.

[18] Estimating uncertainty in elasticity estimates with the same sample used to choose tuning parameters will lead

Table 6: Model Fit Statistics

	Degree of the Polynomial		Out-of-sample
Model	Each Argument (m)	Total ($5m$)	RMSE
NPD-Flex	2	10	0.0176
	3	15	0.0219
NPD-Subst	2	10	0.0489
	3	15	0.0711
BLP			0.0359

selection purposes), we perform five-fold cross validation on the selection sample. Specifically, we randomly select 80% of the weeks in the selection sample, estimate the model on those weeks, and compute the root mean squared error (RMSE) on the remaining 20%. Here, RMSE refers to the square root of the mean of squared ξ errors, where the mean is across store-weeks and products. Repeating this five times and averaging yields a scalar measure of out-of-sample fit.

Table 6 reports predictive RMSEs for each model specification. NPD models are fit with polynomials of degree 10 and 15. We find that the approximation of degree 10 performs best for both NPD-Flex and NPD-Subst. Across model classes, we find that NPD-Flex yields a substantial improvement in fit relative to NPD-Subst and BLP. This provides evidence that allowing for flexible derivatives is useful in this product group, and we will see how this plays out in the estimated elasticities reported in the next section. In light of this, we will use the NPD-Flex model as the "ground-truth" model from which we will simulate demand for the pricing counterfactuals.

7.3 Elasticity Estimates

Table 7 reports the full 5×5 matrix of price elasticity estimates in the Refrigerated Juice product group using the *estimation* sample. We specifically report estimates from the three structural models discussed above: the flexible NPD model (NPD-Flex), the NPD model imposing positive cross-price effects (NPD-Subst), and BLP. For point of comparison, we also report estimates from a log-log model which imposes no economic restrictions.[19] Note that NPD and BLP are non-constant elasticity models and so elasticities vary with prices and quantities. We compute elasticities at the price-quantity values observed in the data and report medians in Table 7. We report the set of elasticity curves (i.e., elasticities as a function of price) for the NPD-Flex model in Appendix E.

There are a few observations to make from the elasticity estimates in Table 7. The first is with respect to the signs of the elasticities. In particular, while all own elasticities are estimated to be negative (as expected), the mix in signs of the cross elasticities looks very different across

to invalid inferences. The specific problem is that the model itself becomes stochastic when it is selected using data, and this source of uncertainty is not accounted for in standard inference procedures (see, e.g., Belloni et al., 2014; Taylor and Tibshirani, 2015). The most common solutions involve sample splitting.

[19] This particular specification of the log-log model is different from the specification used in Section 6 in two ways: (i) the model is estimated using all store-weeks using ZIP code, season, and holiday fixed effects; and (ii) the model includes Hausman IVs to address the potential endogeneity of prices.

Table 7: Elasticity Estimates (Refrigerated Juice)

Model	%Δ in Demand	%Δ in Price: Orange	Lemonade	Fruit	Other	Iced Tea
NPD-Flex	Orange	-5.45 (0.07)	0.06 (0.01)	0.36 (0.08)	0.16 (0.02)	0.35 (0.09)
	Lemonade	-1.25 (0.18)	-1.92 (0.02)	-0.03 (0.05)	-0.36 (0.07)	-0.38 (0.08)
	Fruit	-0.39 (0.28)	0.28 (0.02)	-3.71 (0.05)	0.81 (0.04)	1.34 (0.17)
	Other	-2.36 (0.16)	0.15 (0.01)	1.04 (0.06)	-4.57 (0.11)	0.95 (0.09)
	Iced Tea	1.96 (0.26)	0.57 (0.03)	1.88 (0.20)	1.48 (0.04)	-5.15 (0.27)
NPD-Subst	Orange	-3.50 (0.18)	0.41 (0.04)	0.32 (0.02)	0.55 (0.05)	0.37 (0.01)
	Lemonade	0.46 (0.05)	-2.51 (0.12)	0.20 (0.03)	0.44 (0.03)	0.29 (0.01)
	Fruit	0.43 (0.12)	0.57 (0.01)	-1.81 (0.03)	0.78 (0.01)	0.54 (0.04)
	Other	0.29 (0.13)	0.51 (0.03)	0.46 (0.01)	-3.15 (0.13)	0.48 (0.02)
	Iced Tea	1.59 (0.18)	0.50 (0.02)	0.50 (0.04)	0.66 (0.04)	-2.09 (0.06)
BLP	Orange	-1.95 (0.01)	0.02 (0.00)	0.11 (0.01)	0.01 (0.00)	0.02 (0.00)
	Lemonade	0.16 (0.01)	-1.97 (0.01)	0.12 (0.01)	0.01 (0.00)	0.02 (0.00)
	Fruit	0.13 (0.01)	0.02 (0.00)	-0.81 (0.00)	0.01 (0.00)	0.02 (0.00)
	Other	0.16 (0.01)	0.02 (0.00)	0.12 (0.01)	-1.95 (0.01)	0.02 (0.00)
	Iced Tea	0.16 (0.01)	0.02 (0.00)	0.12 (0.01)	0.01 (0.00)	-1.02 (0.00)
Log-Log	Orange	-3.41 (0.04)	0.27 (0.01)	-0.31 (0.05)	0.16 (0.02)	0.26 (0.03)
	Lemonade	4.00 (0.09)	-1.90 (0.02)	-2.10 (0.12)	-0.76 (0.04)	-0.56 (0.06)
	Fruit	-1.26 (0.05)	0.33 (0.01)	-4.58 (0.06)	0.23 (0.02)	-0.38 (0.03)
	Other	2.37 (0.08)	-0.09 (0.02)	-0.54 (0.10)	-2.00 (0.03)	0.12 (0.05)
	Iced Tea	-0.29 (0.07)	0.12 (0.02)	-0.18 (0.09)	0.05 (0.03)	-2.22 (0.05)

Notes: All models include ZIP code, season, and holiday fixed effects and Hausman IVs. In the NPD and BLP models, bootstrapped standard errors are computed using 100 bootstrap replicates.

models. These differences can in part be explained by differences in each model's constraints, or lack thereof. The NPD-Flex and log-log models place no restrictions on the sign of cross-price effects while the NPD-Subst and BLP models constrain cross effects to be positive. We find that both the NPD-Flex and log-log models produce some negative (and precisely estimated) cross elasticities, suggesting that complementarity is present in this system of goods. The fact that some form of complementarity exists and should be accommodated by the demand model is confirmed by the superior predictive fit provided by NPD-Flex relative to NPD-Subst.

The second observation is with respect to magnitudes. We find that the magnitudes of the estimated elasticities are much larger under flexible demand systems (NPD and log-log) than BLP. In the NPD-Flex model, the largest cross elasticities are -2.36 and 1.96; in the log-log model, the largest cross elasticities are -2.10 and 4.00; in the NPD-Subst model, the largest cross elasticity is 1.59. In contrast, the largest cross elasticity in BLP is 0.16. The attenuation of BLP elasticities is perhaps not too surprising since logit models impose very strong restrictions on cross effects (Berry, 1994). While BLP can in principle admit more flexible substitution patterns, this flexibility comes solely through a model of preference heterogeneity for a subset of product characteristic coefficients.

In practice, it can be difficult to include a rich enough set of characteristics to allow BLP elasticities to meaningfully deviate from logit elasticities. Indeed, even in our BLP model which places random coefficients on product intercepts and price, we still find very "logit-like" elasticity estimates.

We also find that demand is estimated to be more elastic under the NPD models relative to both BLP and log-log models. Differences in the own elasticities can in part be explained by differences in the cross elasticities, as discussed above. In microfounded models, the magnitude of the own effect is tied to the magnitude of cross-effects through properties like Cournot aggregation or diagonal dominance. Therefore, the BLP own elasticities should be much smaller in magnitude if the cross elasticities are also small, and NPD own elasticities should be much larger if the cross elasticities are also large. Even though the NPD and log-log models produce cross elasticities that are similar in magnitude, the NPD own elasticities are on average 1.5 times larger. This is likely due to the log-log model's lack of economic structure: own effects are not tied to cross effects, which is at odds with economic theory and can ultimately lead to issues in subsequent pricing conterfactuals (Anderson and Vilcassim, 2001). In contrast, the underlying structure of NPD models imposes an "accounting rule" whereby own effects must get larger to balance off larger cross effects.

7.4 Demand Boundaries and Omitted Variable Bias

We now investigate biases in estimated substitution patterns when a subset of goods are omitted from the full NPD-Flex refrigerated juice demand system estimated above. In particular, we seek to address the following questions: (i) How large is the bias? (ii) What explains the magnitude and sign of the bias? Answering (i) will let us know whether the bias exists, and answering (ii) can help provide guidance towards the selection of goods to include when estimating demand. For simplicity, we focus on the orange juice demand equation (top row of Table 7) and investigate biases in the own price effect/elasticity. We estimate the following sequence of models:

$$\begin{aligned}\sigma_{\text{orange}}(\delta_{\text{orange}}, \delta_{\text{lemonade}}, \delta_{\text{fruit}}, \delta_{\text{other}}, \delta_{\text{icedtea}}) &\to \sigma^{(4)}_{\text{orange}}(\delta_{\text{orange}}, \delta_{\text{lemonade}}, \delta_{\text{fruit}}, \delta_{\text{other}}) \\ &\to \sigma^{(3)}_{\text{orange}}(\delta_{\text{orange}}, \delta_{\text{lemonade}}, \delta_{\text{fruit}}) \\ &\to \sigma^{(2)}_{\text{orange}}(\delta_{\text{orange}}, \delta_{\text{lemonade}}) \\ &\to \sigma^{(1)}_{\text{orange}}(\delta_{\text{orange}})\end{aligned} \tag{23}$$

which iteratively removes the index for one of the four competing goods from the orange juice demand equation. We take the five-good system as the "true model" and each smaller model as a "misspecified model." We then compute the bias in the own-price effect or elasticity as the estimated effect/elasticity in the misspecified model minus the estimated effect/elasticity in the

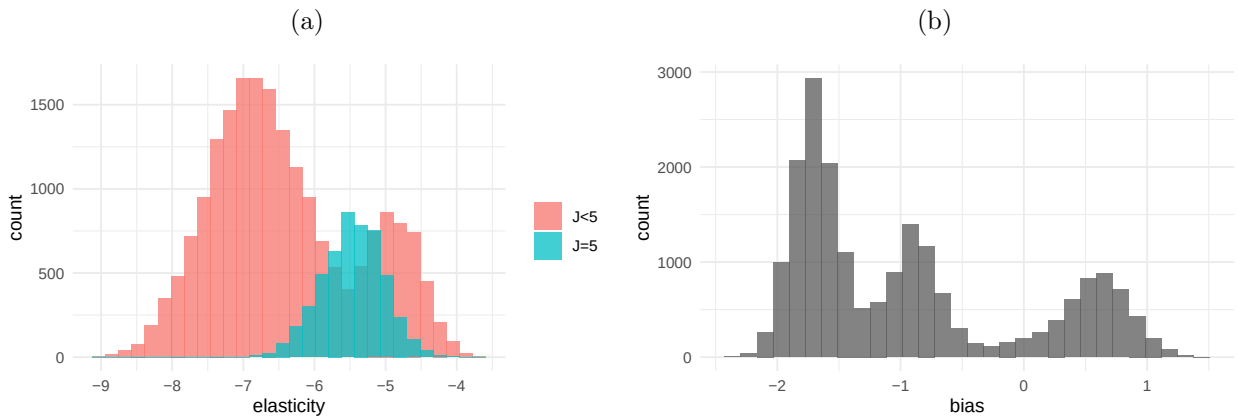

Figure 1: The distributions of orange juice own-price elasticities for the true $J = 5$ model and smaller misspecified models ($J = 1, 2, 3, 4$) are shown in (a). The corresponding distribution of bias is shown in (b). Bias is defined as the estimated elasticity under the smaller misspecified model minus the elasticity obtained from the $J = 5$ model.

true model, i.e. for $J = 1, \ldots, 4$, we compute:

$$\mathcal{B}^{(J)}_{\text{orange}} = \frac{\partial \sigma^{(J)}_{\text{orange}}}{\partial p_{\text{orange}}} - \frac{\partial \sigma_{\text{orange}}}{\partial p_{\text{orange}}} \tag{24}$$

$$\tilde{\mathcal{B}}^{(J)}_{\text{orange}} = \frac{\partial \sigma^{(J)}_{\text{orange}}}{\partial p_{\text{orange}}} \frac{p_{\text{orange}}}{\sigma^{(J)}_{\text{orange}}} - \frac{\partial \sigma_{\text{orange}}}{\partial p_{\text{orange}}} \frac{p_{\text{orange}}}{\sigma_{\text{orange}}} \tag{25}$$

Our analysis below is based on the distribution of either $\mathcal{B}^{(J)}_{\text{orange}}$ or $\tilde{\mathcal{B}}^{(J)}_{\text{orange}}$ across store-weeks.

Are elasticity estimates biased when relevant goods are omitted?

The distributions of orange juice own-price elasticities and bias $\tilde{\mathcal{B}}^{(J)}_{\text{orange}}$ for smaller $J < 5$ models are shown in Figure 1. We find that the distribution of bias ranges from -2.40 to 1.40, with a mean of -0.94 and median of -1.24. Thus, the omission of relevant goods does generate bias and, for the case of orange juice, tends to generate downward (negative) bias. Given a median elasticity of -5.95 for the full model with $J = 5$, an average bias around -1 is substantial and similar in magnitude to bias arising from ignoring the endogeneity of prices in logit models (Nevo, 2001; Chintagunta et al., 2003).

What explains the sign and magnitude of the bias?

In Section 3.3, we reviewed OVB results from a wide class of models including linear models, nonparametric nonseparable models, and models estimated with instruments. In all cases, the sign and magnitude of the bias depends on both the omitted cross-price effect(s) and the relationship between the included prices/instruments and omitted price(s). We now use estimates from the misspecified models in (23) to document the extent to which the same two factors explain the observed bias in our empirical setting.

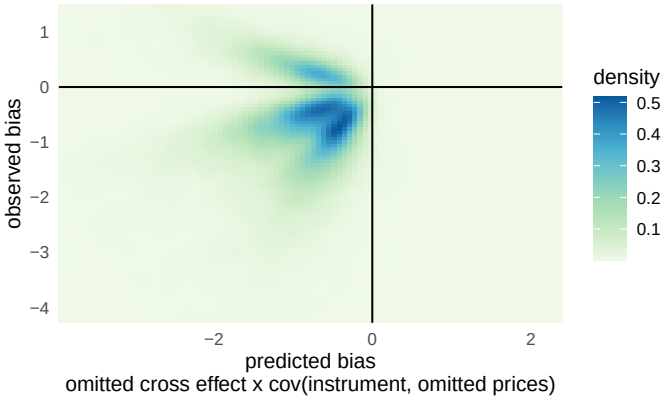

Figure 2: Observed vs. predicted bias for orange juice own-price effects where predictions are made based on (a) only omitted cross-category demand effects or (b) both omitted cross-category demand effects and omitted instrument-price covariances.

Figure 2 shows the relationship between the observed and predicted bias. In particular, for the latter, we use the linear IV formula in (17), which multiplies the omitted cross effect by the relevant instrument-price covariances. The predictions broadly align with the observed biases. In particular, we can correctly predict the sign of 77% of the observations using both the omitted cross effects and instrument-price covariances.

The set of results above suggest that the (albeit naive) predictions using standard OVB formulas can still be useful in understanding the sign of the bias when we omit relevant goods from our nonparametric demand system. At a higher level, our results demonstrate that when we want to understand whether/how elasticity estimates will be affected by the omission of possibly related goods, we should consider both the demand-side substitution between those goods and also the correlation of their prices (or prices and instruments).

7.5 Pricing Counterfactuals

Our final objective is to quantify the costs of misspecified boundaries of product competition through the lens of optimal retailer pricing. We focus on a pricing counterfactual because the first-order conditions associated with the retailer's decision problem directly depend on derivatives of the demand function. As shown in Sections 7.3 and 7.4, such derivatives will be biased when demand equations are misspecified. The costs of misspecification in estimation will also depend on whether a retailer sets prices independently across categories — akin to a retailer pricing under category management (Basuroy et al., 2001) — or jointly across categories, thereby internalizing cross-category substitution. We explore both dimensions of misspecification when solving for optimal retail prices.

First consider the case of "joint pricing" where the retailer sets prices to maximize total profits

across all juice categories.[20]

$$\max_{p_1,\ldots,p_J} \left[\Pi = \sum_{j=1}^{J} \sigma_j(\mathbf{p})(p_j - c_j) \right] \tag{26}$$

Here c_j denotes the marginal cost of category j reported in our data and, with a slight abuse of notation, we use $\sigma_j(\mathbf{p})$ to denote the demand for category j as a function of prices keeping all other demand shifters fixed. The retailer's first-order conditions associated with pricing category j are:

$$\sum_{i=1}^{J} \frac{\partial \sigma_i(\mathbf{p})}{\partial p_j}(p_i - c_i) + \sigma_j(\mathbf{p}) = 0. \tag{27}$$

and so optimal prices crucially depend on a flexible system in order to accurately capture price derivatives $\partial \sigma_i(\mathbf{p})/\partial p_j$.

In contrast, under an "isolated pricing" policy the retailer solves for each p_j^* separately:[21]

$$\max_{p_j} \Big[\Pi_j = \sigma_j(\mathbf{p})(p_j - c_j) \Big] \tag{28}$$

which yields first-order conditions:

$$\frac{\partial \sigma_j(\mathbf{p})}{\partial p_j}(p_j - c_j) + \sigma_j(\mathbf{p}) = 0. \tag{29}$$

The decision problem in (28) is misspecified in the sense that the associated first-order conditions only depend on the derivatives of σ_j and not other related goods in the system. The consequence of isolated category pricing is that the retailer will not be able to internalize the externalities arising from cross-category substitution (Smith and Thomassen, 2012; Thomassen et al., 2017). For example, the NPD-Flex elasticity estimates suggest that a decrease in the price of orange juice will *increase* demand for lemonade, and this complementarity creates a *positive* externality. If the retailer prices jointly across all categories, then it can internalize this externality and the price of orange juice should decrease (relative to a case of isolated pricing). Similarly, our estimates also suggest that a decrease in the price of lemonade will *decrease* demand for orange juice. This substitution creates a *negative* externality and lemonade prices would increase if internalized.

We now have two sources of misspecification: misspecification in the estimation problem and misspecification in the retailer's decision problem. For misspecification in estimation, we use the NPD-Subst model as well as a nonparametric demand model where demand is estimated separately in each category ("NPD-Indep"). We treat the NPD-Flex model as the correctly specified "ground-truth" demand model. For each specification of demand, we then solve for prices under both joint pricing and isolated pricing. In total, this leads to five different configurations of optimal prices (note that NPD-Indep rules out cross-category substitution and so isolated and joint pricing yield the same pricing policy).

[20] We focus on optimization of the baseline prices and keep any discounts (which are observed in the data) fixed throughout the counterfactual exercises.

[21] We assume that the other prices are fixed at the values observed in the data.

Table 8: Profit Table

			Isolated Pricing			Joint Pricing	
		Observed	NPD-Indep	NPD-Subst	NPD-Flex	NPD-Subst	NPD-Flex
Prices	Orange	0.075	0.080	0.087	0.085	0.087	0.085
	Lemonade	0.069	0.069	0.087	0.087	0.122	0.122
	Fruit	0.035	0.035	0.038	0.036	0.038	0.038
	Other	0.067	0.059	0.074	0.066	0.074	0.074
	Iced Tea	0.034	0.038	0.042	0.038	0.042	0.042
Profits	Total	187	202	232	218	265	282

Notes: Medians across store/week combinations. Prices are in dollars per ounce, profits are in dollars.

We solve for optimal prices at the store-week level and report the median prices and profits for a random sample of 1,000 store-weeks in Table 8. First, a few observations about optimal prices. We find that prices tend to be higher under joint pricing, where the magnitude of the difference depends on the degree of substitution estimated by the NPD model. For example, we estimate lemonade to be a strict substitute with all other goods. This negative externality should increase prices when internalized, which is what we find in both NPD-Flex and NPD-Subst specifications. In contrast, consider orange juice where we find that median prices under NPD-Flex are the same irrespective of whether prices are set jointly or not. One explanation is that the NPD-Flex model estimates orange juice to be complements with lemonade and other juices and substitutes with fruit juice and iced tea. These elasticities would induce positive and negative externalities, respectively, which may offset in the pricing problem.

Also note that average optimal prices under the five policies we consider are quite different from average observed prices in the data. We therefore compute a measure of discrepancy between the profit margins predicted by each of our models and the margins observed in the data to test retailer conduct (details on this test and the table of results can be found in Appendix F). We find that the NPD-Subst model with isolated pricing fits best. In fact, all three isolated pricing policies fit the observed margins better than both joint pricing policies, suggesting that the retailer's current pricing practices do not account for cross-category substitution.[22]

Moving to profits, as expected, we find that the joint pricing policy under NPD-Flex is the most profitable, with expected per store-week profits of $282. The differences in profits under the alternative misspecified demand models or decision problems allow us to quantify the cost of misspecification. For example, profits are 6.0% lower when all categories are erroneously assumed to be substitute and prices are solved jointly. Profits are 22.7% lower when demand is correctly specified but prices are solved separately in each category. Profits are 17.7% lower when both the

[22]The discrepancy between joint pricing and the prices observed in the data could occur because the retailer fails to correctly specify the demand model and/or the pricing problem, but it could also be due to the fact that our analysis only considers five out of the many categories of products sold. In other words, the retailer could be internalizing some externalities between these five categories and products outside our demand system. Estimating larger flexible systems could help shed further light on this and we hope the analysis in this paper serves as a useful first step in that direction.

model is misspecified and prices are solved separately in each category. Finally, profits are 28.4% lower when we remove cross-category effects altogether and solve for prices independently in each category. These results suggest that the boundaries of the decision problem are a first-order issue, given that the worst case scenario under joint pricing is still better than the best case scenario under isolated pricing. An important caveat is that our misspecified estimators are still nonparametric and so the profit losses we estimate are likely a lower bound to true losses from pricing with more restrictive parametric models.

8 Conclusion

This paper quantifies the costs of misspecified boundaries of competition on optimal retailer pricing. We explore misspecification in both the estimation problem and the decision problem. Our empirical analysis uses store-level transaction data from a large grocery retailer. We first provide a set of descriptive results across nine diverse product groups to demonstrate that cross-category effects are appreciably large and often hard to sign a priori. We then focus on the Refrigerated Juice product group and estimate demand nonparametrically for all five categories. We also consider more restricted models (based on functional form), as well as smaller models in which subsets of goods are removed from the demand system. We find that the direction of the estimated bias from these smaller misspecified models is largely consistent with standard omitted variable bias results. This suggests that when searching for relevant goods we should consider both demand-side substitution as well as correlations in prices/instruments. Finally, we solve for optimal prices and find that both sources of misspecification lead to profit losses in the range of 6-28%, with misspecification of the decision problem accounting for larger losses relative to misspecification of the demand system.

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APPENDIX

A Additional Simulation Results

We present additional results from the simulation studies in Section 4 with $J = 4$ and $J = 5$. In each case, the true median elasticities are:

$$\mathcal{E}_{jk} = \frac{\partial q_j / q_j}{\partial p_k / p_k} = \begin{cases} -1 & \text{if } j = k \\ -0.15 & \text{if } j = J \text{ or } k = J \\ 0.15 & \text{otherwise.} \end{cases}$$

Below we report the median elasticities across 100 data replicates. Standard errors are given in parentheses.

Table 9: Estimated Elasticities Based on Simulated Data

	%Δ in	% Δ in Price				
J	Demand	1	2	3	4	5
4	1	-0.99 (0.030)	0.14 (0.004)	0.14 (0.005)	-0.15 (0.006)	
	2	0.14 (0.005)	-0.98 (0.026)	0.14 (0.006)	-0.15 (0.006)	
	3	0.14 (0.006)	0.14 (0.005)	-0.99 (0.034)	-0.15 (0.006)	
	4	-0.16 (0.007)	-0.16 (0.005)	-0.16 (0.007)	-1.08 (0.040)	
5	1	-0.99 (0.037)	0.15 (0.006)	0.15 (0.007)	0.15 (0.006)	-0.17 (0.008)
	2	0.15 (0.006)	-0.99 (0.036)	0.15 (0.006)	0.15 (0.006)	-0.16 (0.009)
	3	0.14 (0.006)	0.14 (0.006)	-1.00 (0.036)	0.14 (0.006)	-0.15 (0.007)
	4	0.14 (0.005)	0.14 (0.005)	0.14 (0.005)	-1.00 (0.035)	-0.15 (0.009)
	5	-0.16 (0.007)	-0.16 (0.006)	-0.16 (0.007)	-0.16 (0.006)	-1.13 (0.052)

B Additional Descriptive Results

Figure 3: Price Elasticities and Promotion Effects

C Testing for Consumer Dynamics and Stockpiling Behavior

Hendel and Nevo (2003) propose a descriptive approach to test for the presence of stockpiling behavior using store-level data. The test is based on a regression of quantities ($\log q_{ist}$) on price ($\log p_{ist}$) as well as a variable measuring the duration since the last price promotion. The idea is that, in the presence of stockpiling, consumers will accelerate purchases and buy more during a sale. As inventories deplete weeks after a sale, quantities sold then should increase — i.e., the effect of duration on quantity sold should be positive. We also include controls for feature advertising, a duration variable for feature advertising, and a battery of fixed effects (ZIP code, season, and holiday). We estimate this regression model for each category in each product group. Estimates are reported in Table 10. We find minimal evidence for stockpiling, as the vast majority of estimated duration coefficients are negative and/or very close to zero and precisely estimated.

Table 10: Stockpiling Regressions

		Coefficient			
Product Group	Category	log(Price)	Duration Price	Promotion	Duration Promotion
Baking Goods	FROST	-1.15 (0.038)	0.01 (0.001)	0.33 (0.011)	-0.01 (0.004)
	MIX	-0.92 (0.042)	0.02 (0.005)	0.87 (0.021)	-0.13 (0.010)
Butter/Margarine/Spreads	BTR	-2.14 (0.029)	-0.03 (0.001)	1.37 (0.021)	0.07 (0.002)
	BTRSPRD	-1.73 (0.062)	0.00 (0.000)	0.54 (0.029)	0.01 (0.001)
	MRGRN	0.46 (0.210)	-0.01 (0.002)	1.74 (0.087)	0.04 (0.001)
Canned Fish	CHKLGHT	-0.56 (0.039)	0.02 (0.002)	1.33 (0.017)	-0.02 (0.004)
	CHKWHT	-0.75 (0.035)	-0.01 (0.002)	1.44 (0.027)	0.08 (0.010)
	SALMON	-0.24 (0.010)	-0.01 (0.001)	0.77 (0.034)	-0.04 (0.008)
	SARDINE	0.12 (0.023)	-0.01 (0.000)	0.62 (0.022)	-0.01 (0.008)
	SLDWHT	-0.55 (0.018)	-0.00 (0.003)	0.71 (0.019)	0.04 (0.005)
Cereal	ADULT	-1.54 (0.244)	0.06 (0.002)	1.30 (0.117)	-0.34 (0.010)
	BFY	-0.12 (0.012)	0.00 (0.002)	0.70 (0.012)	0.07 (0.112)
	FAMILY	-1.13 (0.037)	-0.08 (0.007)	1.46 (0.028)	-0.11 (0.240)
	KIDS	-0.30 (0.029)	0.11 (0.003)	2.52 (0.032)	0.15 (0.074)
Crackers	SALTINE	-0.47 (0.039)	0.01 (0.001)	0.56 (0.014)	0.09 (0.008)
	SAVORY	-1.69 (0.054)	0.42 (0.011)	0.39 (0.016)	
	SPRYBTR	-1.52 (0.025)	0.02 (0.003)	0.48 (0.008)	-0.30 (0.054)
Detergent/Softener	DETDRY	-0.39 (0.038)	-0.00 (0.000)		0.02 (0.007)
	DETLQD	-0.24 (0.008)	-0.00 (0.000)		0.09 (0.003)
	SFTDRY	-0.91 (0.044)	-0.01 (0.002)	0.31 (0.009)	0.04 (0.006)
	SFTLQD	-0.98 (0.049)	-0.09 (0.002)	0.49 (0.019)	0.14 (0.006)
Hotdogs	BEEF	-0.63 (0.044)	-0.02 (0.002)	1.48 (0.017)	-0.12 (0.007)
	MEAT	-1.11 (0.094)	0.00 (0.004)	1.38 (0.045)	-0.05 (0.009)
	TRKY	-0.67 (0.053)	-0.02 (0.002)	1.10 (0.027)	-0.13 (0.012)
Jams/Jellies/Peanut Butter	JAM	-1.37 (0.072)	-0.08 (0.002)	0.66 (0.022)	-0.30 (0.017)
	JAMORG	-1.17 (0.060)	-0.00 (0.001)	0.38 (0.019)	-0.08 (0.052)
	PB	-1.72 (0.046)	-0.02 (0.005)	0.86 (0.020)	-0.15 (0.006)
	PBORG	-1.58 (0.064)	0.00 (0.004)	0.81 (0.033)	0.05 (0.063)
Refrigerated Juice	FRUIT	-2.44 (0.129)	-0.00 (0.001)	0.79 (0.041)	0.02 (0.005)
	ICECOFFEE	0.81 (0.062)	0.00 (0.001)	0.55 (0.019)	0.15 (0.012)
	ICETEA	-1.72 (0.043)	0.00 (0.001)	0.36 (0.015)	-0.03 (0.011)
	LEMONADE	-1.70 (0.036)	-0.02 (0.002)	0.27 (0.028)	0.14 (0.013)
	ORANGE	-1.97 (0.049)	0.10 (0.004)	0.91 (0.016)	-0.04 (0.007)
	OTHER	-1.74 (0.055)	-0.01 (0.001)	0.73 (0.035)	0.04 (0.010)

Notes: Standard errors are shown in parentheses.

D First-Stage Price Regressions for Refrigerated Juice

Table 11: Regressions of Prices on Exogenous Characteristics and Instruments

	Orange		Lemonade		Fruit		Other		Iced Tea	
	(I)	(II)	(I)	(II)	(I)	(II)	(I)	(II)	(I)	(II)
Promotion Variables	X	X	X	X	X	X	X	X	X	X
Hausman IVs		X		X		X		X		X
R^2	0.61	0.88	0.35	0.66	0.64	0.79	0.25	0.80	0.86	0.94
F-stat	24.44	92.98	8.66	25.05	28.45	49.35	5.26	51.49	99.19	198.65

Notes: Each regression also includes ZIP code, season, and holiday fixed effects.

E NPD Elasticity Curves

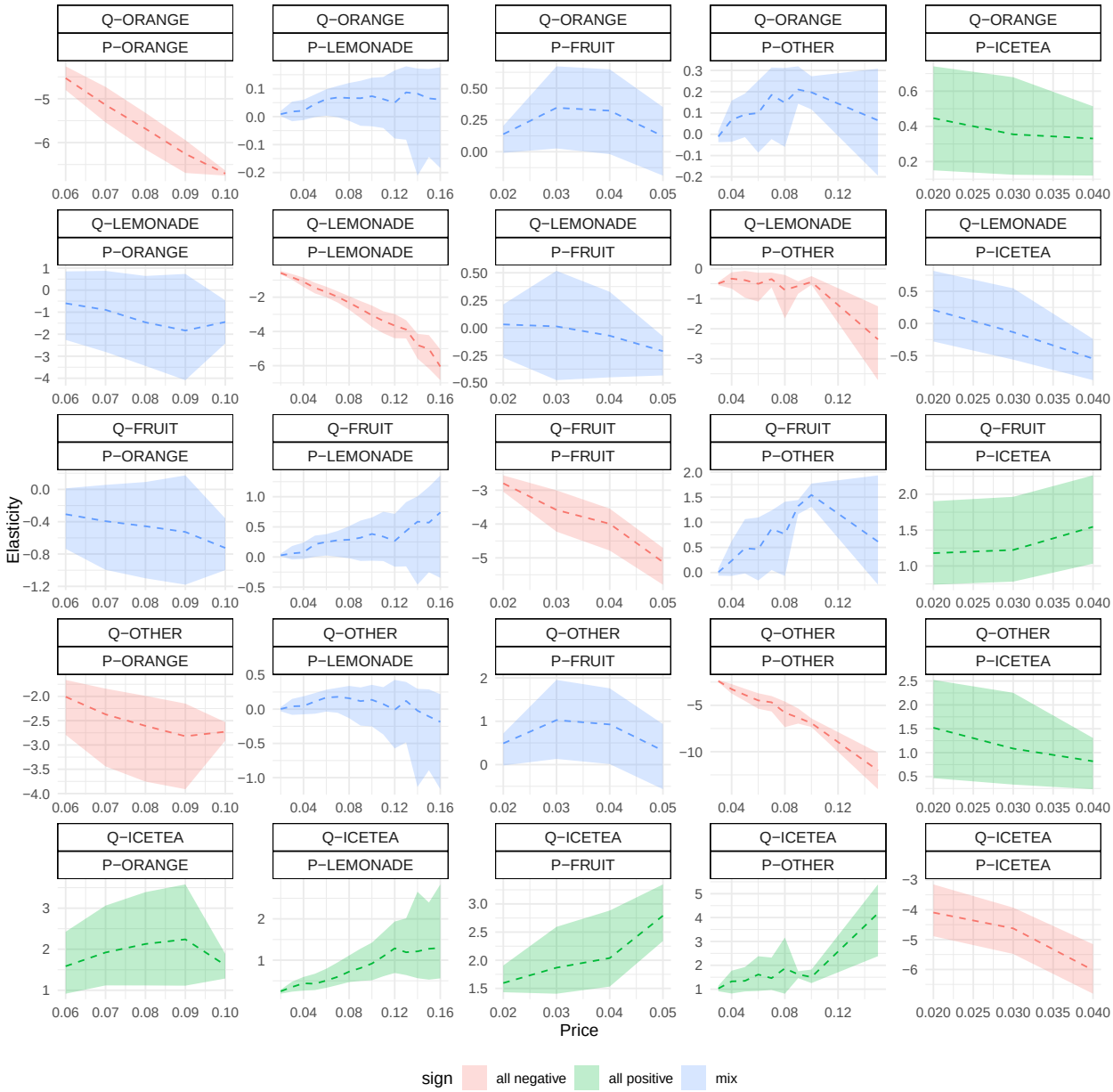

Figure 4: Elasticity Curves (Refrigerated Juice)

F Testing Retailer Conduct

Following Chintagunta et al. (2003), for each model, we compute a distance metric of the form

$$\sum_t \left(\mathbf{m}_{t,data} - \mathbf{m}_{t,model}\right)' W \left(\mathbf{m}_{data} - \mathbf{m}_{model}\right) \tag{30}$$

where $\mathbf{m}_{t,data}$ is the $J-$vector of profit margins in market t, $\mathbf{m}_{t,model}$ is the corresponding vector of margins as predicted by the model, and W is a $J-$by$-J$ weighting matrix. We take W to be the inverse of the correlation matrix of the margins in the data (so that noisier margins receive lower weights than more precisely estimated margins).

The values of this distance metric for different models are reported in Table 12. One can see that the NPD-Subst model with isolated pricing fits the observed margins best, closely followed by the NPD-Flex model with isolated pricing.

Table 12: Test of Retailer Conduct

	Isolated Pricing			Joint Pricing	
	NPD-Indep	NPD-Subst	NPD-Flex	NPD-Subst	NPD-Flex
Distance	21,153	20,547	20,580	30,035	29,496

Notes: The distance metric shown in the table is defined in equation (30).